

BAHub - Phase 13: ERP Mapping & System Alignment

This document maps the modules of the **BAHub** application to functional components of enterprise Resource Planning (ERP) solutions (such as SAP, Oracle, Zoho ERP, or Salesforce).

1. Functional ERP Mapping Matrix

BAHub's architecture mirrors the transactional, relational, and administrative structures of enterprise ERP systems.

BAHub Module	Core ERP Component	Mapping Explanation
Organization Tenant Boundary	Multi-Entity Consolidation	Maps to ERP multi-company setups. Each tenant operates in an isolated environment with separate chart of accounts, users, and resources.
Requirements Split Grid	Product Catalog & BOM	Maps to the Bill of Materials (BOM) in Manufacturing ERP. Requirements act as the structural specifications required to build a final product.
Stakeholder Register	CRM & HR Directory	Maps to Customer Relationship Management (CRM) databases and Human Resources (HR) staffing rosters, logging contact info and interest levels.
Document Generator	Contract & Order Generation	Maps to ERP Order Management or Sales Quote generation. Combines project requirements and data into a document for client sign-off.
Change Request Approvals	Engineering Change Orders (ECO)	Maps to Change Management workflows in Manufacturing/Logistics ERP, tracking, reviewing, and approving modifications to baselines.

Tenant Subscription & Seats	License & Asset Management	Maps to Billing, Receivables, and Asset tracking. Checks seat limits against subscription tiers, restricting usage based on payment terms.
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2. Deep Dive: ERP Module Equivalents

2.1 Sales & Customer Relationship Management (CRM)

In an ERP like Salesforce or Zoho CRM, opportunities move through pipeline stages (Discovery, Proposal, Negotiation, Closed/Won). In BAHub:

- **The Opportunity Pipeline** maps to the **Requirements Split Grid**. Requirements transition through statuses (`DRAFT` , `REVIEW` , `APPROVED`) as agreement is reached.
- **Customer Contacts** map to the **Stakeholder Matrix**. Stakeholders are logged with Power/Interest ratings, helping PMs focus alignment efforts on key drivers.

2.2 Operations, Workflows, & Approval Engines

Enterprise ERPs use strict workflows and approval logic (such as SAP Workflow or Oracle Approvals Management) to prevent unauthorized transactions. BAHub implements equivalent controls:

- **Change Request Approvals:** When a BA submits a `ChangeRequest` , the ticket is locked in `REVIEW` status until a Product Owner or Admin reviews and approves it.
- **Document Sign-off:** The PO reviews compiled BRDs/FRDs and signs off. This logs signature metadata (`signed_off_by` , `signed_off_at`) and freezes the version, establishing a baseline.

2.3 Finance & Subscription Billing

In a financial ledger, transaction balances must balance. In BAHub, subscription states must map directly to tenant usage limits:

- **Billing Tiers:** The `TenantSubscription` model links to the organization context and maps subscription levels (`FREE` , `PRO` , `ENTERPRISE`).
- **Seats Limit Enforcer:** When inviting users, a service counts active user records against the subscription seat limit (e.g. 5 users limit on the `FREE` tier), blocking invitations if the limit is exceeded.

2.4 Warehouse & Inventory (Backlog Tracking)

In a logistics ERP, inventory items are tracked in storage lanes and checked for stock shortages. In BAHub:

- **Storage Lanes** map to the **Kanban Agile Backlog**. User story cards represent inventory units moving through status lanes (`TODO` , `IN_PROGRESS` , `QA` , `DONE`).
 - **Stock Shortage Checks** map to **Jira Backlog Synchronization**. A celery runner checks if user stories are synced to Jira, updating the database status with Jira issue keys to prevent backlog drift.
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3. Dashboards & Consolidated Analytics Reports

ERP systems use analytical reports to help executives monitor performance and manage operations. BAHub implements equivalent dashboards:

- **Executive Dashboard**: Consolidates active project metrics, requirement distributions, story point completion, and risk levels.
- **Risk Heat Matrix**: Maps risks by probability (High, Medium, Low) and impact (High, Medium, Low) on a visual grid, helping PMs prioritize mitigation tasks.
- **Jira Sync Audit Logs**: The `SyncLog` database table acts as a ledger, logging the status (Success/Failed), timestamp, and trigger details for all integration runs.